23125

C23-C–302

# 23125

## BOARD DIPLOMA EXAMINATION, (C-23)

## MARCH/APRIL—2025

## DCE – THIRD SEMESTER EXAMINAT00ION

MECHANICS OF SOLIDS AND THEORY OF STRUCTURES

*Time* : 3 Hours ] [ *Total Marks* : 60

**PART—A** 3×10=30

**Instructions :** (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

**1.** Define *(a)* Flexural rigidity *(b)* Neutral axis

**2.** A simply supported beam 100 mm wide, 200 mm deep and 3m long is carrying u.d.l. of 30 kN/m over the entire span. Determine the maximum Shearing Stress.

**3.** A Simply Supported Beam of 3·5m span carries a central point load of 300 KN. If the value of *EI* is $4 \times 10^{13}$ N-mm$^2$, find maximum Deflection at center.

**4.** Distinguish between strength and stiffness of a beam.

**5.** Define *(a)* Effective/ equivalent length *(b)* Slendereness ratio.

**6.** A Concrete dam of rectangular section 15 meters height retains water with a free board of 3 meters. Find the total pressure of water on one meter length of the dam.

**7.** Define *(a)* Active earth pressure *(b)* Passive earth pressure.

**8.** A cantilever of span 6m carries a point load of 15 kN at 2 m from fixed end. If the beam is propped at the free end. Determine the magnitude of the prop reaction.

**9.** State the merits and demerits of continuous beams.

**10.** Define a frame and list the classification of frames.

## PART—B

10×5=50

**Instructions :** (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

**11.** A teak wood beam is to be simply supported over a span of 5 m. It carries a u.d.l. of kN/m over entire span. Find the dimensions of the cross-section of the beam of bending stress is not to exceed 8 $N/mm^2$. The depth of the beam is be twice the breadth.

**12.** An I-section with rectangular ends has the following dimensions. Flanges 200 ×20 mm, web 300 ×20 mm and total depth 340 mm. Find the maximum shear stress developed in the beam for a shear force of 50kN. Also sketch the shear stress distribution across the section.

**13.** A simply supported beam of span 6 m carries a central point load of 25 kN and UDL of 10 kN/m, throughout. Find the maximum slope and deflection if E=210 $kN/mm^2$ and I= 320 $\times 10^3$ $mm^5$ .

**14.** A cantilever of 4 meters length carries a UDL of 10kn/m for a length of 4 meters from fixed end and a point load of 12 kN at the free end.Determine maximum slope and deflection at the free end by moment-area method. Given E = 200 kN=mm$^2$ and I = 320 × $10^5$ mm$^4$.

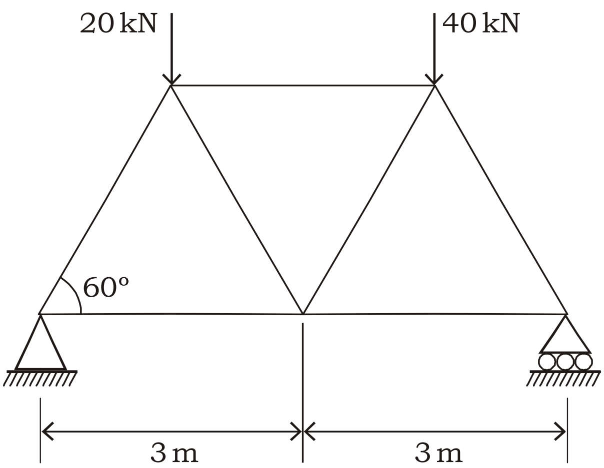
**15.** A 2m long steel tube is used as a column with both ends fixed. The internal and external diameters are 4 cm and 5 cm respectively. If the Rankine's constants are 320 N/mm$^2$ and 1/7500, determine the ratio of Euler's and Rankine's loads. Also suggest the safest load a factor of safety = 5 and E = 2 ×$10^5$ MPa.

**16.** A masonry dam 8m high. 1·5m wide at top and 5m wide at bottom retains water to a depth of 7·5m. The water face of dam is vertical.Find the maximum and mimimum stresses at base. Weight of masonry 22·4 kN/m$^3$ and Specific weight of water 10 kN/m$^3$

**17.** A horizontal cantilever beam 5 meters long carries a point load of 12·5 kN at 2 meters from fixed end.If the beam is propped at free and to the level of fixed end.Find the load on the prop a0nd construct S.F and B.M diagrams.

**18.** Find the forces in any five members of the given truss analytically by method of joints.

Take all angles as 60°

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